JIAZHI HE

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EDUCATION

University of Pittsburgh *Aug 2023 - Present*
Ph.D. in Statistics

University of Science and Technology of China (USTC) *Sep 2019 - Jul 2023*
B.S. in Mathematics and Applied Mathematics

RESEARCH INTERESTS

I have broad research interests in methodology and theory of high-dimensional statistical inference, dynamic treatment strategies, sequential multiple assignment randomized trial (SMART), adaptive designs, causal inference, win statistics for multiple endpoints and statistical machine learning, to establish reliable, powerful, and interpretable solutions to wide real-world problems.

RESEARCH EXPERIENCE AND WORKING PAPERS

Recent Advances in the Analysis and Design of Sequential Multiple Assignment Randomized Trials to Improve Patient Outcomes
University of Pittsburgh, Advisor: **Prof. Yu Cheng, Prof. Abdus S. Wahed** *June 2025 - Present*

- A review paper to be submitted to WIREs computational statistics.

Weighted Win Odds Regression for Sequential Multiple Assignment Randomized Trials
University of Pittsburgh, Advisor: **Prof. Yu Cheng** *Mar 2025 - Present*

- Proposed a doubly-weighted U-statistic-based Generalized Estimating Equation (UGEE) to estimate model coefficients, addressing challenges of censoring and "missingness" from sequential randomization.
- Derived a large-sample variance using Hájek projections of the doubly-weighted U-statistic kernel.
- Validated the estimator via extensive Monte Carlo simulations, confirming it achieves nominal 95% coverage.

Weighted Composite Endpoint for DTRs in SMARTs with Competing Risks
University of Pittsburgh, Advisor: **Prof. Yu Cheng** *Oct 2025 - Present*

- Developed a weighted composite endpoint framework for SMARTs by combining competing risks with stakeholder-elicited weights; modeled the composite counting process via a semiparametric proportional-rates model.
- Derived the estimating equation and Breslow-type baseline estimator, and implemented a Newton–Raphson solver in R.

Optimization for Perinatal Lifestyle Interventions, HABIT SMART Trial
University of Pittsburgh, Advisor: **Prof. Yu Cheng** *Jan 2025 - Sep 2025*

- Applied Q-learning methods to evaluate optimal dynamic treatment regimes (DTRs) in a longitudinal behavioral intervention study (HABIT), targeting improvements in postpartum weight, cardiometabolic health, depressive symptoms, and stress.

- Developed and validated regression models in R to derive an optimal dynamic treatment regime, revealing that ideal treatment pathways varied significantly by race and engagement with the prenatal intervention.
- Confirmed the effectiveness of the data-driven treatment rules, showing that adherence to the optimal regime resulted in a significant reduction in postpartum depression symptoms.

SELECTED PROJECTS

Statistical Consulting, University of Pittsburgh *Sep 2024 - Dec 2024*

- Evaluated the impact of genotype, age, and strain on muscular dystrophy progression in mouse models. Conducted 2-way and 3-way repeated measures ANOVA and linear mixed-effects modeling to analyze grip strength and muscle contractility data.
- Provided data-driven recommendations to the University of Pittsburgh's Music Department by analyzing ordinal survey data on student satisfaction with facilities. Utilized descriptive statistics and visualizations, and proposed advanced methods including the Chi-Square test of independence and proportional odds models with random effects.

Analysis of HIV Drug Resistance Data, University of Pittsburgh *Dec 2023*

- Developed and executed a robust feature selection pipeline using a multivariate linear regression model to identify HIV-1 mutations associated with drug resistance across seven PI-type drugs.
- Constructed and evaluated a final predictive linear model for APV resistance using features selected via forward stepwise selection based on the Bayesian Information Criterion (BIC). The model's validity was confirmed using a 70/30 train-test split.

Optimization Algorithms Course Project, USTC *May 2022*

- Established the model of an Assignment Problem and obtained the optimal solution using Ortools on Python.
- Modeled and solved a Traveling Salesman Problem based on optimization software Gurobi and implemented Branch&Cut algorithms; Compared the effect of Lazy Cut and User Cut on the solution rate.

Hybrid CNN for Amazon Star Rating Prediction, USTC *May 2021*

- Engineered the foundational components of a Convolutional Neural Network (CNN) from scratch, including the manual implementation of the forward and backward propagation algorithms for both convolutional and fully-connected layers.
- Developed a hybrid model to predict Amazon customer star ratings (1 to 5) based on text comments, utilizing Bert pre-trained word vectors as input features.
- Designed a two-stage network architecture that initially used linear regression (CNN-based) and then added classification layers to optimize for extreme values (1-star and 5-star reviews).

SKILLS

Programming Languages and Frameworks
C, Python, R, LaTeX, Mathematica, Matlab, SQL

Languages
Mandarin (Native Speaker), English (Fluent), German (College German Band 4)
GRE General Test: 324 = Verbal (155)+Quantitative (169)+ Analytical Writing (3.5)

SELECTED AWARDS AND SCHOLARSHIPS

- **Outstanding Volunteer Award**, USTC *2021*
- **Outstanding Student Scholarship, Silver Award**, USTC *2020*
  Second-class departmental honor, winning rate < 10%.
- **National High School Mathematics League, Second Prize in Anhui Province** *2017*

TEACHING

Teaching Assistant/Fellow at University of Pittsburgh

- STAT 1000: Applied Statistical Methods *Spring 2025, 2024, Fall 2025*
- STAT 1060: Data Science Foundations *Fall 2024*
- STAT 1201/2200: Applied Nonparametric Statistics *Fall 2023*
- STAT 0800: Statistics in the Modern World *Fall 2023*

Teaching Assistant at USTC

- MATH1009: Linear Algebra B1 *Spring 2022*
- MATH3005: Mathematical Statistics *Fall 2022*